

RESEARCH METHODOLOGY

CSC4321: SPECIAL TOPICS IN COMPUTER SCIENCE



OUTLINE

- **RESEARCH OVERVIEW**
- **RESEARCH PROPOSAL**
- **LITERATURE REVIEW**
- **CITATION AND REFERENCING**
- **RESEARCH PRODUCTIVITY TOOLS**
- **REPORT WRITING, PUBLISHING AND PRESENTATION**
- **CRASHED COURSES**
- **GROUP PRESENTATION TOPICS –**

(Trends in Computer Science)



RESEARCH OVERVIEW

1.1 Definition

1.2 Types of Research

1.3 Scope of a Research

1.4 Attributes of a High-quality research

1.5 Research Ethics

1.5.1 Allocation of credit in publications

1.5.2 Research originality

1.6 Research Hypothesis



1.1 Definition

- Research can be defined as a human activity based on application of intellect in the investigation of a solution to a problem; it involves gathering of data, information and facts for the advancement of knowledge.
- Formal research is a structural method; it is usually taken to be concerned with the systematic development of new knowledge. It primarily starts with the searching of existing knowledge and formulation of hypothesis. Different methods are used for different kinds of researches. The choice of a method depends on what the method can do, how it can be applied, and what kind of data is accessible.
- Researches in the field of information technology (IT) are holistic in nature, owing to the fact that they bring IT solutions for numerous fields. IT researches also proposes new research methods, provide new IT tools and discover new ideas.



1.2 Types of Research

- Basically, we have two types of research; Basic and Applied Research. Basic Research is also known as fundamental, pure or academic research. It is driven by scientists' interest in a scientific question. The main motivation is to expand man's knowledge, not to create or invent something. There is no obvious commercial value to the discoveries from basic research.
- For example, basic science investigations probe for answers to questions such as How did the universe begin? or What are protons, neutrons, and electrons composed of?
- Applied research is designed to solve practical problems of the modern world, rather than to acquire knowledge for knowledge's sake. One might say that the goal of the applied scientist is to improve the human condition.
- Example of questions asked in applied research: How can the lead emission emanating from PCs be reduced? How can we cure cancer? How the time complexity of the binary search algorithm be improved? How can we use communication technologies to prevent examination malpractice?
- There are many instances when the distinction between basic and applied research is not clear. However, the difference lies in the foreseeable practical application. Some scientists feel that the time has come for paradigm shift from purely basic research to applied research.
- We could see other research types such as Descriptive, Observational, Survey, Case study, and Correlational Research etc.



1.3 Scope of a Research

- We could also categorize research based on scope. The scope of a research process is to update an existing knowledge or to produce some new knowledge. This, in principle, can take three main forms:
- Exploratory research: also known as feasibility study, here a new problem can be structured and identified. It also helps in identifying the possibility of undertaking research and deciding worthy area to be researched. For instance, developing a computer that is independent of the current PLs.
- Constructive research: a (new) solution to a problem can be developed. For instance, developing a new PL.
- Empirical research: empirical evidence on the feasibility of an existing solution to a problem can be provided. For instance, investigating the efficiency of one PL over another.



1.4 Attributes of a High-quality research

Benefiting a society is the ultimate aim of a good research. A good research has the following attributes:

- It is based on previous work: this does not mean plagiarism, but always looking at work done by others to provide a basis for what and how a new work can be done.
- It is replicable: research is an activity that can be replicated. It makes researcher's argument stronger.
- It is generalizable to other settings: if replicated based on a standard, the outcomes should be in conformity.
- It is based on some logical rationale and tied to theory: provides answers to the questions that help fill-in pieces to what can be a large and complicated puzzle.
- It is doable: it has a conceptual framework.
- It is cyclical in nature: answering today's scientific questions leads to tomorrow's scientific questions.
- It is incremental: research is incremental.
- It is an apolitical activity.



1.5 Research Ethics

Ethics are of paramount importance at every stage of research, such as planning, conducting and evaluating a research project. At first, research ethic involves considering the potential cost and benefits of the research, and should avoid data breach, and unnecessary risk of harming people, or property. For this reason, Researcher should avoid deception on other stakeholders and should obtain informed consent from all involved in the study.

As an IT researcher, it is advisable to pay attention to the following:

- Do not use your proficiency to hack the client's system illegally. Always preserve privacy and confidentiality of your clients.
- Do not possess a pirated stuff, or plagiarize the work of others and do not skew results to solicit extra reward.
- Do not commit science fraud, falsify research or otherwise conduct scientific misconduct.
- Do not use your position as a peer reviewer or system tester to give fake reviews that will punish or damage fellow scientists.
- Basically, research must follow all regulations given, and anticipate possible ethical problems.
- Remember, whistleblowing is one mechanism to help discover misconduct in research.
- IT researcher ought to have vast knowledge outside of science, to include business, social studies, and languages among others. With narrow focus on the basics of IT, you will make an amateur PC operator.



1.5.1 Allocation of credit in publications

- The principle of fairness and the role of personal recognition is of great emphasis for proper allocation of credit. In scientific publications, credit is explicitly acknowledged either in the list of authors, in the acknowledgments section, and in the list of references or citations. Conflicts over proper attribution can arise in any of these places. Citations serve many purposes in a scientific paper. Failure to cite the work of others can give rise to more than just hard feelings.
- As an IT researcher, the number of days, months or years spent on a product should not tempt you to assure completeness. You should bear in mind that your byproduct is an outcome of a provisional finding. Meaning, you may not always provide a complete accurate solution in-line with predefined user requirements. This is because all scientific results must be treated as susceptible to error and acceptable limitations. As such, be ready to maintain and upgrade your findings and byproducts. Else, the truth will likely emerge as problems arise in persisting investigations. Beyond honest errors are those that involve deception; fabrication, falsification, and plagiarism.



1.5.2 Research originality

- Originality is an important measure of a high-quality research. Developing and implementing necessary skills to tell original and non-original research is a must for any potential researcher. Originality is a multidimensional measure; it could be based on the research method and materials used, based on the data and results obtained, or based on the report written.
- A measure of originality based on ideas and facts could be as follows:
 - New facts + New ideas (100% original)
 - New facts + Old ideas
 - Old facts + New ideas
 - Old facts + Old ideas (0% original)
- It is clear that, 100% original and 0% original research have the highest level of new findings and highest level of plagiarism, respectively. Both could have high rejection rate. This is because 100% original is too original and it might repel appreciation as it lacks conformity with existing facts and ideas. While a 0% original research is scientifically unethical and should not be counted as a research.



1.6 Research Hypothesis

- A hypothesis is an assertion that has not (yet) been either well supported or ruled out by experiment. Generally, a hypothesis is used to make predictions that can be tested by observing the outcome of an experiment. If the outcome is inconsistent with the hypothesis, then the hypothesis is rejected. However, if the outcome is consistent with the hypothesis, the experiment is said to support the hypothesis.
- Technically, the word hypothesis is slightly a mathematical term for “sentence” or “claim” or “statement”; a statement that something is true concerning particular population. For instance, “CBT reduces students’ score” is a hypothesis that could be tested based on available data. How would you test this hypothesis?



RESEARCH PROPOSAL

2.1 What is Research Proposal?

2.2 Structure of a Research Proposal

2.3 Purpose of Writing Proposal



2.1 What is Research Proposal?

- Proposals are meant to enable readers assess the feasibility of a research, there by persuading readers to decide on doing something about the research (i.e. accepting or funding the research). Readers assess the feasibility of a research based on how well your proposal answer the following questions: WHAT you are proposing? HOW you plan to do it? WHEN you plan to do it? and HOW MUCH it is going to cost?
- Investing sufficient time and thought into writing a proposal document will yield a good return and can save you a lot of time, confusion and disappointment when actually embarking on a research or project. NOTE: In proposal document factual information are preferred to promotional information.



2.2 Structure of a Research Proposal

- By producing a well-structured proposal, you are jumping ahead toward the final report. This is because it provides key sections similar to those in the final report.
- The three main sections of a proposal document are:
 - Introduction
 - Literature Review/ Background Information
 - Methodology/ Procedure (including Time and Budget)
- The introduction section tells the reader what the project is all about, why the project is worth doing, and why your project is a good topic.
- Review section is where you demonstrate to your reader your knowledge on the research topic. Knowledge of researches done on the topic, the current state of the art, and tools and techniques to be mastered for the research. More importantly, how your research approach builds on or vary from previous researches.
- Procedure is the heart of your proposal because it depicts the way you proposed to carry out the research. Methods on how research tools will be used, what data collection and analysis techniques to be applied. In essence, it gives a future narration on the final 'research methodology' section. In extension, it categorically states all tasks, subtasks, resources and scheduling required in the work (usually using Gantt chart).



2.3 Purpose of Writing Proposal

Proposal document tells what your main aim is, your planned method of achieving it and what are your needs (time and money) and perhaps who needs your work.

- It gives an overview of the relevance and objective of a research project.
- It gives an overview of the content, the procedure and the timing of a research project.
- It shows whether a research project is manageable in scope and time.



LITERATURE REVIEW



- 3.1 What is a Literature Review?
- 3.2 Characteristics of a good Literature review
- 3.3 Hints on Ensuring the Quality of Literary Sources
- 3.4 Purpose of Literature Review

3.1 What is a Literature Review?

- Any kind of academic research starts with a literature review and contains a literature review as a section. This is because it generally updates readers or researchers on current trends, history, and possible future dimensions based on facts, methods and/or standards.
- Literature review can be viewed as either a research activity, a research section or a research paper. As a section in a write-off, it is a body of text that aims at reviewing the key points of current knowledge on a particular topic. While literature review as an article purposely interpret and/or assess works in a particular subject area. Thus, it is either a pointer to information sources or a comprehensive exploration on the views of the key contributors in a field.
- It traces the intellectual progress and major debates in a field by summarizing and synthesizing the most relevant information. It also provides new and combined interpretations on old and new materials respectively.



3.2 Characteristics of a good Literature review

- Its flow of ideas has to be logical either thematically, chronologically or methodologically.
- It contains current and relevant references, cited consistently and appropriately.
- Adhere to proper use of terminologies.
- It did not support any of existing contradicting views, but evaluates sources and advise readers on the most pertinent or relevant of the views.



3.3 Hints on Ensuring the Quality of Literary Sources

Quality of a review largely depend on the credibility of sources. Nowadays, with abundant information on the internet it is very difficult to judge the credibility of an online paper. Quick and accurate evaluation of credible source can only be achieved through experience, yet these few tricks might help:

1. Before going to the internet, begin with an academic library.
2. Scan abstract and introduction sections of a paper; this will guide you in determining its quality and relevance to your research.
3. Learn how to check the credentials of any source.
4. Always remember, few credible sources are far preferable than a long list of irrelevant sources.
5. Look for agreements in research results but never reject a research just because it conflicts with your hypothesis.



3.4 Purpose of Literature Review

- To identify research gaps
- To avoid repeating what was achieved and mistakes made by others
- To build on the platform of existing knowledge and ideas
- To identify researchers with similar research interest and build research network
- To enhance knowledge of your research interest
- To identify useful works in your area
- To put your work into perspective by establishing its intellectual context; positioning it relative to other works
- To identify opposing views
- To demonstrate your ability to access previous works in an area
- To identify facts, methods and results that could be relevant to your work.



CITATION AND REFERENCING

4.1 What is Citation and Referencing?

4.2 Format and Styles of Citation and References

4.3 Samples of Citation and References



4.1 What is Citation and Referencing?

- The main purpose of this course is for you to learn how to effectively carry out a research on virtually every subject, to make standard documentation and presentation; showing your sources of information.
- While documenting, you might use information from textbooks, manuals, encyclopedia, and other published articles. Citation and referencing is the standard way of showing your sources of information which allow readers to easily locate the source or trace it back.
- A writer cites the sources of information while writing and provide details about the sources during referencing. Details on a source explains what kind of source it is and where readers could locate it if necessary. In essence, citations are done within the write-off while referencing came at the end of a write-off as a detailed list of the citations.
- Many standards/ styles of citation and referencing are in existence and they vary across academic fields. In this course we pay more attention to American Psychological Association (APA) citation style. This style is commonly used for publishing in the social and natural sciences. Primarily, citations in the body of the text includes name of the author and date of publication in parentheses. If the author's name was just mentioned in the text, it is not necessary to repeat it.



4.2 Format and Styles of Citation and References

- In-text Citation for a single author, has the author's last name and year separated by a comma. For instance: (Ahmad, 2015). Else, for at most three authors, use their last names separated by commas and with an ampersand "&" before the very last name in the list, then the year separated by a comma. For instance: (Ahmad & Idris, 2016) or (Ahmad, Sani, Ali & Idris, 2016). Else, for at least six authors, use the first author's last name and "et al." For example: (Ahmad, et al., 2017).
- References usually come toward the last page of an article. To create a list of references, one for each item cited in the paper. The references are to be alphabetized by the first authors' last name. If you cite more than one paper by the same first author, sort them by year of publication.
- Single-space the entries in your list of references. Start at the left margin for the first line of each bibliography entry. Each additional line of each entry should be indented a reasonable amount. Separate the entries with a blank line. Do not number the references. Doing so means you have to renumber all the references whenever you insert a new reference.



4.3 Samples of Citation and References

- *Citations Within text:*

Number of Authors	At the beginning of a sentence	At the end of a sentence
Single author	Ahmad (2014)	(Ahmad, 2014)
At most three authors	Ahmad and Cavus (2015)	(Ahmad & Cavus, 2015)
At least four authors	Ahmad et al., (2014)	(Ahmad et al., 2014)
Website	Model Checking Contest, (2015)	(Model Checking Contest, 2015)



4.3 Samples of Citation and References

■ **Reference Section:**

Journal

Ahmad A.L., Cavus, N. (2015). Motives behind Preference of Internet Communication Tools among University Students. *International Journal Science and Engineering Research*, 36, 301-321.

Book

Ahmad A.L. (2017). IT Research and Project Management. Kano, Nigeria: Kano Printing Press.

Book Chapters

Zakari A., Ahmad A.L., Girish B. (2017). Towards improving the security of low-interaction honeypots. In D. Treagust, D. Duit, and B. Fraser (Eds.), *Lecture Notes in Electrical Engineering*. (pp. 131-141). Springer, Cham.

International Conference

Ahmad A.L., Cavus, N. (2015). Efficacy of employing social media for educational practices in the unrest regions of the world. In *Proceedings of the 4th Cyprus International Conference on Educational Research* (pp. 363-368). Cyprus: Girne American University.

Web Sites

Model Checking Contest, France. (2015). Model of Internet Open Trading Protocol (IOTP model) using Petri Net. Retrieved June 19, 2016 from <http://mcc.lip6.fr/models.php>



RESEARCH PRODUCTIVITY TOOLS

- 5.1 Scientific Databases
- 5.2 Citation and referencing tools
- 5.3 Plagiarism checkers
- 5.4 Grammar checkers
- 5.5 Paraphrasing tools



Intro

- Research Productivity Tools are software packages that help in making research activities much more manageable, while helping researchers meet research expectations. Various tools are employed in research formulation, data collection/analysis, literature review, report writing, proofreading and so on. In the following subsections, we will look at few of the exemplary Research Productivity Tools with class demonstrations.



5.1 Scientific Databases

- Research databases are organized collections of computerized information or data such as periodical articles, books, graphics and multimedia that can be searched to retrieve information. Databases can be general or subject oriented with bibliographic citations, abstracts, and or full text. Exemplary academic research databases include Scopus, Web of Science, PubMed, ERIC, and IEEE Xplore



5.2 Citation and referencing tools

- Citation management tools - sometimes called citation managers, reference management tools, or reference managers - enable you to store, organize, annotate, and share your references for research. They can also be used with a word processor to automatically insert in-text citations and format bibliographies according to your desired style, such as APA, Chicago, MLA, and many more. Exemplary citation and referencing tools include EndNote Basic, Mendeley and Zotero.



5.3 Plagiarism checkers

- The widespread use of computers and the advent of the Internet have made it easier to plagiarize the work of others. Plagiarism detection or content similarity detection is the process of locating instances of plagiarism or copyright infringement within a work or document. Plagiarism checkers are intelligent software tools for detecting duplication from your content online against billions of web pages. Exemplary plagiarism checkers include iThenticate, Turnitin and other software packages such as Grammarly that have plugins for plagiarism detection.



5.4 Grammar checkers

- Grammar checkers are writing assistants that not only corrects grammar but also provide comprehensive writing feedback. For instance, Grammarly helps in proofreading and checking for grammatical errors and spelling errors in social media updates, emails, blog posts, and articles.



5.5 Paraphrasing tools

- A paraphrasing tool, also known as a content spinner, article rewriter, or sentence rephraser has a single task: to paraphrase your content to the desired level of uniqueness while maintaining the original meaning. Exemplary paraphrasing tools include Spinbot and QuillBot.



REPORT WRITING, PUBLISHING AND PRESENTATION

6.1 Structure of Scientific Reports - IMRAD

6.2 Different Types of Research Methodologies (Case study, survey, observations, experimental etc.)

6.3 Presentation and Publishing



6.1 Structure of Scientific Reports - IMRAD

- **Introduction**
- **Background and Pre-writing**
- **Writing Processes**
- **The Basic Structure – IMRaD**
- **Publishing Processes**



Introduction - Assumptions

- ❑ We are familiar with basics of research, types, scope, and attributes of a high-quality research
- ❑ Research Ethics, the need for allocation of credit in publications, research originality, research hypotheses
- ❑ Research Proposal and Proposal Writing
- ❑ Literature Review
- ❑ Research Productivity Tools
- ❑ Scientific Databases **et. al.**



Pre-writing



Background and Pre-writing

- *You did an experiment or research study for your work, and now you have to write it up for experts in the field to review and young scientists to learn.*
- *You feel that you understood the background sufficiently, designed and completed the study effectively, obtained useful data, and can use those data to draw conclusions about a scientific process or principle.*



Background and Pre-writing

cont'd

- *But how exactly do you write all that? What are their expectations? How can you display your findings in a decent way? etc. etc.....*
- *Generally speaking, people investigating some scientific hypotheses have a **responsibility to the rest of the scientific world** to report their findings, particularly if these findings add to or contradict previous ideas.*



Background and Pre-writing

cont'd

As you can probably imagine, people reading such reports have two primary goals:

- ✓ ***They want to gather the information presented.***
- ✓ ***They want to know if the findings are legitimate.***

Your job as a writer, then, is to fulfill these two goals.



Background and Pre-writing

cont'd

Before Starting to Write

- **The best way to prepare to write a paper is to make sure that you fully understand everything you need to about the research.**
- **Obviously, if you don't quite know what went on during the research process, you're going to find it **difficult to explain it satisfactorily to someone else.****



Objective

By the end of this session we should be able to:

- **Discuss the purposes of writing and publishing research papers**
- **Plan our paper publication**
- **Understand the structure of a good paper**
- **Best practice of sourcing information for your publication**
- **Organise your information**
- **Use an appropriate style of writing**
- **Present data and findings effectively**
- **Understand how to communicate in the right journal and appropriate way to handle reviewers and editors**



The Importance of Research Paper Writing

- **Advancing Knowledge and groundbreaking research.**
- **Career Advancement and Academic Recognition.**
- **Funding Opportunities and Grants.**

All these came from well-written research papers.

~Publish or Perish~



Plan our paper publication -

The need to understand Your Audience

- **Identifying the Target Readership.**
- **Tailoring Language and Style for Different Audiences**
- **Considering the Needs and Expectations of Reviewers; understanding what reviewers look for can improve your chances of publication.**



Plan our paper publication –

The need to select a Research Topic

- **Criteria for Choosing a Research Topic are shaped by relevance, feasibility, and personal interest.**
- **Finding Gaps in Existing Literature can lead to novel contributions.**
- **A good hint is to balance Personal Interest for a topic with Research Trends.**



Plan our paper publication -

The need to conduct Literature Review

- **The Purpose and Goals of a Literature Review.**
- **Strategies for Effective Literature Search using databases, search engines, and other tools.**
- **Literature Synthesis.**



Plan our paper publication –

The need to Define Objectives and Questions

- **Crafting Clear and Specific Research Objectives can set the direction for your study, e.g., "To assess the impact of X on Y in Z population."**
- **Formulating Research Questions and Hypotheses to guide the investigation and offer testable hypotheses.**
- **Aligning Research Goals with Significance.**



Plan our paper publication –

Identify Research Methodology

- **Choosing the Appropriate Research Method from surveys to experiments, and so on.**
- **Data Collection Techniques and Tools such as questionnaire items, interviews questions, ICTs or apparatus for laboratory experiments.**
- **Addressing Ethical Considerations in Methodology esp. when working with human subjects or sensitive data.**



Plan our paper publication –

Understand our Data Collection and Analysis

- **Data Collection Procedures and Challenges** such as data quality, sample size, and data collection biases. These can significantly affect research outcomes.
- **Data Cleaning and Preprocessing** is often necessary for raw data before analysis, using methods like missing data handling, and normalization.
- **Statistical Analysis Methods** entail mastering the data distribution, employing the appropriate technique and tools; descriptive, inferential, qualitative & quantitative analyses, depending on your research type.
- **Packages** – SPSS, MATLAB, Excel etc.



Plan our paper publication –

Understanding the importance of Accurate Citations

- **Accurate citation is the key to giving proper credit to previous work and avoiding accusations of plagiarism.**
- **Using research productivity tools such as Mendeley, Zotero and understanding citation styles (APA, MLA, Chicago, etc.); and adhering to standards for in-text citations and reference list entries.**
- **There are always plagiarism checks and originality metrics against the common avoidable pitfalls.**



Writing Processes



Consider the following steps:

1. Ask yourself: Why would we benefit from this knowledge?

Answering the question will lead you to a more complete understanding of the research you have carried out, and this "big picture" will in turn help you write a successful report.

2. Whenever you can't answer a question regarding your research make use of your supervisor. He/she will probably be able to explain it to you (or, at least, help you figure it out).



Goal & Objective Setting



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Basic Structure of Your Research Paper

- The format, sometimes called "IMRaD," (Introduction, Methods, Results [and] Discussion) may take slightly different shapes depending on the discipline or audience; some ask you to include an abstract or separate section for the hypothesis, or call the Discussion section "Conclusions" or change the order of the sections (some professional and academic journals require the Methods section to appear last).
- IMRaD is the typical structure used in scientific papers. **What each section should contain?** Subsections and Headings do improve the organization of a well-structured paper.
- Emphasis should be put into Flow and Logical Organization of Content inter/intra sections, naturally leading to the next.



IMRaD

Section	Scientific method step	As well as...
Introduction	States the hypothesis of your work	explains how you derived that hypothesis and how it connects to previous research; gives the purpose of the experiment/study
Methods	details how the hypothesis was tested	clarifies why you performed your study in that particular way
Results	provides raw (i.e., un-interpreted) data collected	expresses the data in table form, as an easy-to-read figure, or as percentages/ratios
Discussion	considers whether the data obtained support the hypothesis	explores the implications of your finding and judges the potential limitations of your experimental design



IMRaD –

Crafting an Effective Introduction

- **Hooking the Reader with a Strong Opening** entails strategies like starting with a surprising fact, a thought-provoking question, or a relevant anecdote to engage the reader.
- **Defining the Research Problem** by clearly articulating the problem your research addresses.
- **Stating the Research Objectives** and explain its broader Significance.
- **The readers of academic journals** tend to be professionals in the field, authors explain the background in order to permit readers to evaluate the study's pertinence for their own work.



IMRaD –

Developing a Strong Methodology

- **Detailing the Research Design - to describe your research design with clarity, and structure.**
- **Explaining Data Collection and Analysis Procedures.**
- **Strong Methodology always assures Replicability/Reproducibility and Validity.**
- **As with any piece of writing, your Methods section will succeed only if it fulfills its readers' expectations, so you need to be clear in your own mind about the purpose of this section. In this section, you want to describe in detail how you conducted your research and clarify the rationale for your procedure.**
- **In science, it's not sufficient merely to design and carry out a research.**



IMRaD

Content of Methodology

- Sometimes the hardest thing about writing this section isn't what you should talk about, but what you shouldn't talk about. Writers often want to include the results, because they measured and recorded the results during the course of the process (i.e. experiment). But such data should be reserved for the Results section.
- In the Methods section, you can write *that* you recorded the results, or *how* you recorded the results (*e.g.* in a table), but you shouldn't write *what* the results were, not yet. Here, you're merely stating exactly how you went about testing your hypothesis.
- How much detail? Be precise in providing details, but stay relevant.



IMRaD

Ask yourself, "Would it make any difference if this piece were a different size or made from a different material?" If not, you probably don't need to get too specific. If so, you should give as many details as necessary to prevent this experiment from going away if someone else tries to carry it out. Probably the most crucial detail is measurement; you should always quantify anything you can, such as time elapsed, temperature, mass, volume, etc.



IMRaD

Rationale: Be sure that as you're relating your actions during research, you explain your rationale for the protocol you developed. In a professional setting, writers provide their rationale as a way to explain their thinking to potential critics. On one hand, of course, that's your motivation for talking about protocol, too. On the other hand, since in practical terms you're also writing to your supervisor, explaining the rationale indicates that you understand the reasons for implementing the research in that way, and that you're not just following orders. Critical thinking is crucial, robots don't make good scientists.

NOISE TO SIGNAL
RobCottingham.com



Siri's heartbreaking legacy



IMRaD

b. Structure and style

Organization is especially important in the Methods section because readers must understand your procedure completely. Many writers are surprised by the difficulty of conveying what they did during the research process, since after all they're only reporting an event, but it's often tricky to present this information in a coherent way. There's a fairly standard structure you can use to guide you, and following the conventions for style can help clarify your points.

- Subsections: Occasionally, researchers use subsections to report their procedure when the following circumstances apply:**
 - 1) if they've used a great many materials;**
 - 2) if the procedure is unusually complicated;**
 - 3) if they've developed a procedure that won't be familiar to many of their readers.**



IMRaD

Narrative structure: Think of this section as telling a story about a group of people and the experiment they performed. Describe what you did in the order in which you did it.

Past tense: Remember that you're describing what happened, so you should use past tense to refer to everything you did.



IMRaD –

Presenting Results and Findings

- **Data Visualization using Tables and Figures to effectively present data, and highlight key findings.**
- **Design and Formatting well-designed tables and figures.**
- **Visuals aids enhance reader's Comprehension and Retention of the most complex dataset.**
- **There is great implication in Interpreting and Discussing Key Findings. ~You just have to look for every key metric and get it right~.**
- **Situate your paper within the literature by Supporting metrics with Evidence and Citations.**



IMRaD –

Presenting Results and Findings

- **The Results section is often both the shortest and most important part of your report. Your “Materials and Methods” section shows how you obtained the results, and your “Discussion” section explores the significance of the results, so clearly the Results section forms the backbone of a scientific report.**
- **This section provides the most critical information about your performance: the data that allow you to discuss how your hypothesis was or wasn't supported. But it doesn't provide anything else, which explains why this section is generally shorter than the others.**
- **Before you write this section, look at all the data you collected to figure out what relates significantly to your hypothesis.**



IMRaD –

Presenting Results and Findings

a. Text

This should be a short paragraph, generally just a few lines, that describes the results you obtained. In a relatively simple case, one that doesn't produce a lot of data for you to repeat, the text can represent the entire Results section. Don't feel that you need to include lots of extraneous detail to compensate for a short (but effective) text; your readers appreciate discrimination more than your ability to recite facts. In a more complex case, you may want to use tables and/or figures to help guide your readers toward the most important information you gathered. In that event, you'll need to refer to each table or figure directly, where appropriate. Feel free to describe trends that emerge as you examine the data.



IMRaD –

Presenting Results and Findings

b. Tables

When you do have reason to tabulate material, pay attention to the clarity and readability of the format you use. Here are a few tips:

Number your table. Then, when you refer to the table in the text, use that number to tell your readers which table they can review to clarify the material.

Give your table a title. This title should be descriptive enough to communicate the contents of the table, but not so long that it becomes difficult to follow. The titles in the sample tables above are acceptable.



IMRaD –

Presenting Results and Findings

c. Figures

How to include figures in the report? Although tables can be useful ways of showing trends in the results you obtained, figures (i.e., illustrations) can do an even better job of emphasizing such trends. Report writers often use graphic representations of the data they collected to provide their readers with a literal picture of how the experiment went.

Figures can include maps, photographs, drawings, flow charts, bar graphs, and section graphs ("pie charts").



IMRaD –

Discussion and Interpretation

- **The discussion section is probably the least formalized part of the report, in that you can't really apply the same structure to every type of research effort. In simple terms, here you tell your readers what to make of the “Results” you obtained. If you have done the “Results” part well, your readers should already recognize the trends in the data and have a fairly clear idea of whether your hypothesis was supported. Because the “Results” can seem so self-explanatory.**
- **Basically, the “Discussion” contains several parts, in no particular order, but roughly moving from specific to general:**
 - **Explain whether the data support your hypothesis**
 - **Acknowledge any anomalous data or deviations from what you expected**
 - **Derive conclusions, based on your findings, about the process you're studying**
 - **Relate your findings to earlier work in the same area (if you can)**
 - **Explore the theoretical and/or practical implications of your findings**



IMRaD –

Discussion and Interpretation

- **Interpreting Results in the Context of Research Questions – link your findings back to your research questions and objectives.**
- **Addressing Limitations and Potential Bias on how they might impact the research and give chance for future research agenda.**
- **Formulating Conclusions and Implications drawn from your research findings, and explain their relevance to who, when, where and why.**



IMRaD – CR

Conclusions And Recommendations

- **This is the section of the report which draws together the main issues.**
- **It should be expressed clearly and should not present any new information.**
- **You may wish to list your recommendations in separate section or include them with the conclusions.**



Writing the Abstract

- **Purpose and Elements of an Abstract.** Abstract's role in summarizing the study. Key elements – objectives, methods, results, and conclusions.
- **There is always tips for Writing a Concise, Informative and well-structured Abstract within Word Limit and Graphical Abstract.**
- **There are variations between journals on Abstract Length and Style. Adhere to the journal's style guidelines.**



Crafting an Engaging Title

- **Characteristics of a Good and Engaging Research Paper Title include elements of clarity, conciseness, and informativeness.**
- **Capturing the Essence of the Study to effectively encapsulate the research's focus.**
- **Strike a balance between Clarity and Creativity.**





search ID: aba0682

***"I DID NOT WRITE THE REPORT,
I DOWNLOADED AND PRINTED IT DIRECTLY
FROM THE INTERNET, BUT I DID COLLATE AND
STPLE IT MYSELF."***



Publishing Processes



The Peer Review Process

- Overview of Peer Review in Academic Publishing
- Benefits of Peer Review
- Responding to Reviewer Comments
- Editor and Reviewer Interactions
- Manuscript Handling and Review Times
- Acceptance, Revision, or Rejection



Preparing Your Manuscript for Submission

- **Formatting Guidelines for Different Journals**
- **Pre-submission Checklist**
- **Manuscript Submission/Handling Platforms**
 - **ScholarOne Manuscripts**
 - **Editorial Manager**
 - **Open Journal Systems (OJS)**
 - **Manuscript Central**
 - **Evise**
 - **Susy et al.**



Choosing the Right Journal

- Criteria for Journal Selection
- Matching Research Scope and Audience
- Impact Factor and Reputation

<https://journalfinder.wiley.com/>

<https://journalsuggester.springer.com/>

<http://cofactorscience.com/journal-selector>

<https://journalfinder.elsevier.com/>

et al.



Open Access vs. Traditional Publishing

- **Exploring Publication Models**
- **Advantages and Disadvantages of Open Access**
- **Copyright and Licensing Considerations**

Promoting Your Research

- **Strategies for Research Visibility**
- **Leveraging Social Media and Academic Networks**
- **Collaborating with Institutions and Authors**



Common Pitfalls to Avoid

- **Common Mistakes in Research Paper Writing**
- **Lessons from Rejections and Revisions**
- **Proactive Measures for Avoiding Pitfalls**

Tips for Effective Proofreading

- **Proofreading Techniques and Strategies**
- **Utilizing Proofreading Tools**
- **Seeking Peer and Professional Review**



Ethical Considerations in Publishing

- Avoiding Plagiarism and Self-Plagiarism
- Ethical Handling of Data
- Authorship and Acknowledgments



6.2 Different Types of Research Methodologies

(Case study, survey, observations, experimental etc.)



RESEARCH METHODS

1. Experimental Research
2. Natural Observation
3. Survey
4. Case Study
5. Correlational Research
6. Simulation and Modelling
7. Integrated Research



"Perfection of *means* and confusion of *goals* seem, in my opinion, to characterize our age."

Einstein, 'Out of My Later Years'



1. Experimental Research

- **Experimental research methodology involves a systematic investigation where researchers manipulate variables to observe their effects, establishing cause-and-effect relationships with the aim of testing hypotheses, validating theories, and contributing to the understanding of phenomena. For instance, in computer science we employ experimental methods to validate algorithms, evaluate system performance, or assess user experiences.**
- **Experimental methods involve Independent variables (manipulated) vs. dependent variables (observed) with consideration on control groups for isolating the impact of the independent variable and implementation of random assignment to minimize bias.**
- **Different experimental data collection strategies ensure precision and reliability in line with the research goals. Data collection tool might include surveys, sensors, or software logs with established protocols for consistent and accurate data recording and documentation practices.**



- **Statistical analysis using techniques such as t-tests, ANOVA, regression analysis is employed for hypothesis testing and result interpretation. The results interpretation is commonly based on the p-values as a pointer to results' significance.**
- **Statistical tools like R, Python with libraries (NumPy, SciPy), or dedicated software for analysis such as SPSS are used to run the data analyses.**
- **Results are presented using either descriptive (e.g. mean, median, and mode) or inferential statistics (making predictions or inferences).**
- **Some of its weaknesses stem from the possibility of biases such as selection and experimenter biases. Best practice is to identify and mitigate the biases. In addition to ethical considerations related to participant consent, confidentiality, and the responsible use of technology.**



2. Natural Observation

Natural Observation is the basic method of getting into about any event. It becomes a scientific tool for research when observation is used in a systematic manner with a scientific attitude. It implies accurate watching of phenomena as they occur in nature with regard to cause and effect relationship. In observation the researcher goes to places where the event is taking place and record the findings.



2.1. Observational Research

Observational research consists of systematic observation. The word *systematic* implies a sensible and replicable procedure for collecting data. Data might be collected by taking notes or with a camera recording device or by administering a questionnaire or in any other way that can be adequately described for other scientists who wish to repeat the observations.



No experiment is conducted. The researcher does not attempt to alter the world during the data collection phase. The data are analyzed, and researchers look for interesting or important patterns.

The naturalistic observation can be defined as a non-experimental approaches used in the field or in real-life settings. In the naturalistic observation method the researcher very carefully observes and records some behavior or phenomenon, sometimes over a prolonged period, in its natural setting.

The subjects or phenomena are not directly interfered with in any way. In the natural sciences this may involve observing an animal or groups of animals or some physical phenomena, such as the eruption of a volcano *etc.*



Observational studies can provide answers to “what” phenomena occurred, particularly when people are involved in a process, along with gaining insights into “why” the phenomena occurred. There is a great tendency especially in software engineering research towards observational studies. Usually there is an effort to minimize the observer's impact by carrying out observations, passively, secretly or from a hidden vantage point. i.e. with preschoolers a stationary camera with a wide-angle lens can be put near the ceiling in one corner of the room, without influencing behavior.



2.2. Controlled observation (*Systematic observation*).

Controlled observation occurs when observational research is carried out under carefully arranged conditions. Each subject is exposed to the same situation, to see differences between individual reactions.

For example, a group of babies may be exposed, one at a time, to a laboratory situation called *stranger approaches*.



A stony-faced stranger approaches the mother and takes the baby out of her arms.

Around the age of 7-8 months, some babies seem to find this very alarming. Some cry even at the sight of an approaching stranger. However, not all babies go through this "stranger anxiety" phase.

Researchers can test hypotheses about what factors make stranger anxiety more or less likely by testing a variety of mother/baby pairs. The controlled setting and carefully arranged conditions allow researchers to compare reactions of different children.



2.3. Microanalysis

Microanalysis is detailed analysis of very brief events. Sometimes researchers notice interesting things simply by *slowing down a quick movement*. This can be done with video cameras set to capture many images per second. For example, studying the hunting techniques of a wild species.

2.4. Standardized testing

Standardized testing is a form of controlled observation using testing procedures previously shown to be reliable and valid.



2.5. Strengths

The major strength of this method is that it allows researchers to observe behavior in the setting in which it normally occurs rather than the artificial and limited setting of the laboratory. Further uses might include studying nature for its own sake or using nature to validate some laboratory finding or theoretical concept.

2.6. Limitations

The limitations of this method are many.

First and foremost this is a descriptive method, not an explanatory one. That is, without the controlled conditions of the laboratory, conclusions about cause-and-effect relationships cannot be drawn. Behavior can only be described, not explained.



This method can also take a great amount of time. Researchers may have to wait for some time to observe the behavior or phenomenon of interest.

Further limitations include the difficulty of observing behavior without disrupting it and the difficulty of coding results in a manner appropriate for statistical analysis.



3. SURVEY

- **TV stations rely on surveys to profile the people who watch programmes over their stations.**
- **Automobile companies use survey to find out about customer satisfaction with the cars they produce or sell.**
- **Statistics Departments conduct surveys to measure the consumer price index with the purpose of advising policy.**
- **Engineering companies conduct surveys gathering information about the habits of road users.**
- **Manufacturing companies conduct surveys to determine consumer acceptance of their products.**



WHAT IS A SURVEY?

The survey, a type of non experimental, descriptive study, does not involve direct observation by a researcher. Rather, inferences about behavior are made from data collected via interviews or questionnaires. Interviews or questionnaires commonly include an assortment of forced-choice questions (e.g. True-False) or open-ended questions (e.g. short answer essay) to which subjects are asked to respond.



Survey research is one of the most important areas of measurement in applied social research. The broad area of survey research encompasses any measurement procedures that involve asking questions of respondents. A "survey" can be anything from a short paper-and-pencil feedback form to an intensive one-on-one in-depth interview.

This sort of data collection is sometimes referred to as a self-report. Once again, this is a non experimental, descriptive approach.



3.1. Strengths

Surveys are particularly useful when researchers are interested in collecting data on aspects of behavior that are difficult to observe directly (such as thoughts about political choice) and when it is desirable to sample a large number of subjects.

Surveys are used extensively in the social and natural sciences to assess attitudes and opinions on a variety of subjects.

3.2. Limitations

This method is descriptive, not explanatory, and, therefore, *cannot offer any insights into cause-and-effect relationships.*

The major limitation of the survey method is that it relies on a self-report method of data collection. Intentional deception, poor memory, or misunderstanding of the question can all contribute to inaccuracies in the data.





GOOD EVENING SIR, I'M DOING A SURVEY
ON HOME SECURITY.

From: <http://www.cartoonstock.com/directory/s/survey.asp>



4. Case Study

It involves obtaining detailed information about an individual event (natural or artificial) and/or to develop general principles about behavior.

Case study research excels at bringing us to an understanding of a complex issue or object and can extend experience or add strength to what is already known through previous research.

Case studies emphasize detailed contextual analysis of a limited number of events or conditions and their relationships. Researchers have used the case study research method for many years across a variety of disciplines. It is sometimes very helpful to study one event (natural or artificial) or to study one person (or a very small group of people) in great depth to learn as much information as possible. This method is particularly useful in studying rare disorders or circumstances.



Streghths

- Case studies are particularly useful when researchers want to get a detailed contextual view of an individual's life or of a particular phenomena.
- In the social sciences they are often used to help understand the social and familial factors that might be part of the development of some form of deviant behavior in an individual.
- Natural scientists might use this method to study a single animal or a single instance of some physical phenomenon.
- Case studies are also useful when researchers cannot, for practical or ethical reasons, do experimental studies.



Limitations

- **First and foremost this is a descriptive method, not an explanatory one. That is, without the controlled conditions of the laboratory, conclusions about cause-and-effect relationships cannot be drawn. Behavior can only be described, not explained.**
- **Case studies also involve only a single individual or just a few and therefore may not be representative of the general group or population.**
- **In the social sciences case studies often rely on descriptive information provided by different people. This leaves room for important details to be left out. Also, much of the information collected is retrospective data, recollections of past events, and is therefore subject to the problems inherent to memory.**



6.3 Presentation and Publishing

- No matter how hard you've tried throughout your research process, the impact of your findings solely relies on the level of your scientific achievement, how well you present it, and prestige of the academic database that publish your work.
- After ascertaining the purpose of your presentation (to inform, to persuade, to inspire, or to teach), you should have at the back of your mind the key constraints related to your audience.
- Moreover, the determinant of how you should communicate with the audience relies on their professional status, knowledge of the topic, and their assessment role on your presentation.
- A coherent presentation involves active interaction with the audience by making use of your voice, movement and stage presence. During slide presentations you should draft the outline (beginning, middle and end) and slide alignment (horizontal or vertical) to be used. Take note of the approximate timing and what information to include or exclude.

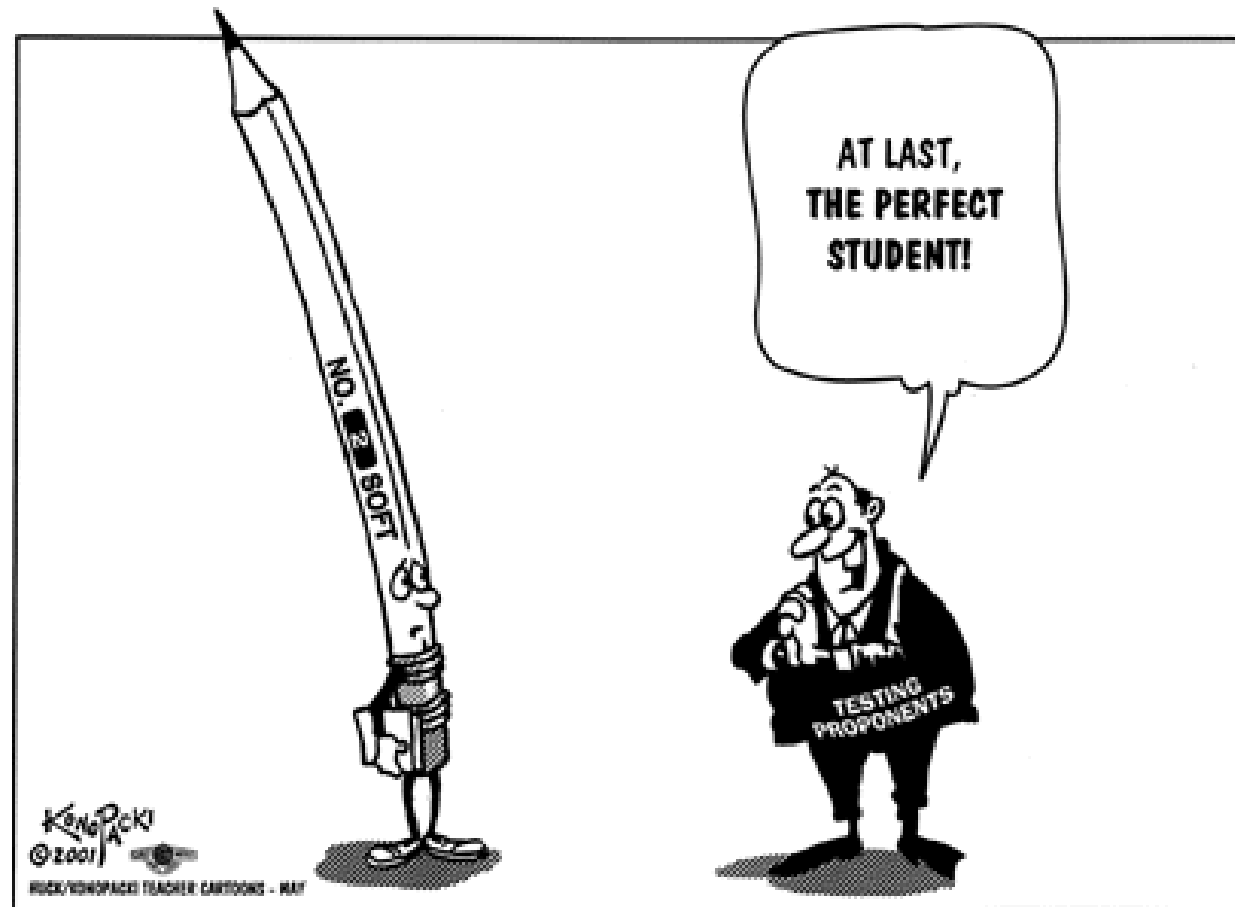


6.3 Presentation and Publishing ...

You may adopt any of the following presentation styles:

- Memorizing the Speech: it allows eye contact. But it is difficult for long speeches, provide room for precision errors and has no room for improvising.
- Reading from a Text: it ensures precision. But its flow does not sound natural, no room for improvising and hinders eye contact.
- Winging It: it sounds natural but has higher room for error.
- Speaking from Slides: it insures organization, allows eye contact and allows improvising. But has some room for error.





From: <http://www.solidarity.com/hkcartoons/teachertoons/miketesting2.html>



CRASHED COURSES

7.1 Overview of Internet and Web Programming

7.2 Overview of Programming with C++, Java and Python

7.3 Database



7.1 Overview of Internet and Web Programming

- **Basics of Internet** (Internet and WWW, Internet, Intranet and Extranet, Internet Service Providers, Internet Connections, Web page, Website, Portal, Hyperlink, Web servers, Web Browsers, Search Engines, URL, IP Address)
- HTML tags and attributes
- Text formatting
- Adding multimedia contents
- Creating tables
- Creating forms
- Site map and linking pages
- CSS
- JavaScript



7.2 Overview of Programming with C++, Java and Python

7.2.1 Introduction to C++: Syntax, Keywords, Output, Comments, User Input

7.2.2 Variables and Data Types: Variables Declaration (Rules and syntax), Data Types (integer, char, float, double)

7.2.3 Operators and Control Flow Statements

7.2.4 JAVA as covered in OOP

7.2.5 Python Introduction

What is Python?

What can Python do?

Why Python?

Python Syntax compared to other programming languages

Python Getting Started

Python Install

Python Quickstart

The Python Command Line



7.3 Database



GROUP PRESENTATION TOPICS

**TEN GROUPS
(GROUP 1 – 10)**



Trends in Computer Science

8.1 VR/AR

8.2 Generative AI

8.3 AI/ML

8.4 IoT

8.5 Simulation

8.6 Universal design

8.7 Big data

8.8 3D printing

8.9 Cyber security

8.10 Blockchain

**8.11 Educational technologies
(LMS, MOOC, CBT etc.)**

**8.12 Satellite internet (Starlink
as case study)**

**8.13 e-commerce/ business/
retail**

8.14 Robotics

**8.15 Embedded systems
(Raspberry Pi, Arduino etc)**

8.16 Cloud computing

**8.17 FinTech (Opay, Moniefy,
and other mobile wallets and
payment APIs as case studies)**

8.18 eHealth

**8.19 Social Networking Sites
(Facebook, TikTok, X etc as
case studies)**

**8.20 Dynamics of Data
Storages (emphasis on
memories esp. SSD, and other
innovative data storage media)**

